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2005 Mathematical Contest in Modeling (MCM) Summary Sheet

(Attach a copy of this page to each copy of your solution paper.)

Type a summary of your results on this page. Do not include the name of your school, advisor, or team members on this page.

Summary

This paper is aimed to determine the optimal number of tollbooths design in a toll plaza, and relieve the congestion as possible.

If arriving rate of the plaza entrance does not exceed Peak Capacity of the exit, the whole process that vehicles pass the toll plaza can be divided into two phases. Phase 1 is the queue and the payment, which is a standard M/M/c queueing system. Phase 2 is the departure from the toll plaza. Considering each lane in the exit as a server and Peak Capacity of the exit as average service rate, Phase 2 can also be regarded as an M/M/c queueing system. Based on Queueing Theory, we obtain the total average delay which is consist of the average delay in the two queues. The optimal goal is to minimize the total delay in the plaza.

If arriving rate of the plaza entrance exceeds Peak Capacity of the exit, the queueing model has no efficiency. Based on fluid dynamics, we analyze the relation between traffic flow and velocity. Then our optimal goal is to maximize the leaving vehicles per unit time.

Sequentially, we take two typical toll plazas in Garden State parkway as examples. Applying our model, we confirm the optimal numbers of the tollbooths: 3 tollbooths for Cape May Toll Plaza and 12 for Union Toll Plaza.

To generalize our model, we discuss the toll plaza with Mixed-type tollbooths, and propose two improved methods in designing of a toll plaza system.

Optimal Design of Toll Plaza

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Introduction

With the increasing population density and industrial base in New Jersey, highways such as the Garden State Parkway, interstate 95 experience severe traffic jams at toll plazas even during non-rush hours. The interruption caused by the toll plazas and the traffic jams bring great inconveniences to the public. How to relieve the congestion problems is concerned with the whole society.

A sketch map of the toll plaza is shown in Figure 1. When arriving at the entry,

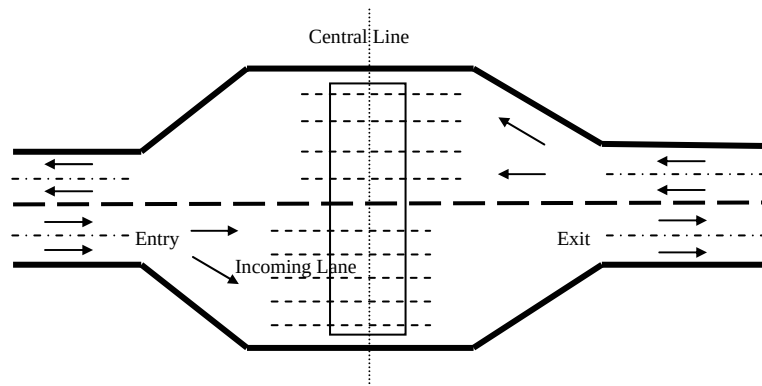


Figure 1
Sketch Map of a Toll Plaza

the flow of vehicles fans out to the incoming lanes, and when leaving the toll plaza, the flow funnels back down to the travel lanes. If the tollbooths are too many, congestion will increase upon departure from the toll plaza normally. On the contrary, if the number of tollbooths is too small, congestion builds at the entry because of the service time. A good design can improve these problems significantly. There are a number of important issues in the toll plaza design problem. Some of them that we concern about are

- › How to characterize the traffic intensities.
- › How many types of tollbooths that we have.
- › What is the traffic delay as a function of the number of tollbooths.
- › What is the optimal goal.

Based on the Queueing Theory, we reduce the whole process that vehicles pass the toll plaza to a twice-queueing problem, and then obtain the delay as a function of the number of tollbooths. Considering the motorist annoyance, we take the average delay time as the optimal goal. However, when the entrance arriving rate is larger than the exit overall Peak Capacity, the method based on Queueing Theory can't work, but it is the very time when congestions take place easily. We analyze the relation between traffic flow and velocity based on fluid dynamics, and this time our optimal goal is to maximize the leaving

vehicles per unit time.

Terms and Notations

Base Capacity (BC): The Base Capacity (cars per hour per lane) of a freeway facility.

Peak Capacity (PC): The capacity of the segment.

A list of notations commonly used is in Table 1.

Table 1
Notions Used.

Notation	Definition	Unit
$X(t)$	the vehicles that reach the toll plaza in the interval of time $[0, t)$	<i>Cars/s</i>
λ_1	mean arriving rate of vehicles	<i>Cars/s</i>
λ_2	mean service rate of each tollbooth	<i>Cars/s</i>
λ_3	the equivalent service rate of each lane in the exit	<i>Cars/s</i>
u	the service time	<i>s</i>
W_q	average vehicles delay in the tollbooth	<i>s</i>
L_q	the length of the queue in the tollbooth	<i>Cars</i>
W'_q	average vehicles delay in the exit	<i>s</i>
L'_q	the length of the queue in the exit	<i>Cars</i>
v	the velocity of the vehicles	<i>m/s</i>
a	the acceleration of the vehicles	<i>m / s²</i>
s	distance between the tollbooths and the exit	<i>m</i>
f_{ID}	Interchange Density	<i>Cars/mile</i>
N	the number of lanes in one direction	
n	the number of tollbooths	
f_{HV}	adjustment factor for heavy vehicles	
f_P	adjustment factor for drive populations	
P_T	proportion of trucks and buses in the traffic stream	
E_T	passenger-car equivalents; 1.5 for all urban highways	

Assumptions

- › There is only one type of payment in the same toll plaza. A mixed-type collection will be discussed in Further Discussion.
- › Considering the velocity, vehicles are divided into two types.
- › There is exactly one tollbooth per incoming lane.
- › Only one direction of the traffic flow is considered.

Preparation

When entering the toll plaza, the motion of vehicles can be described as Figure 2.

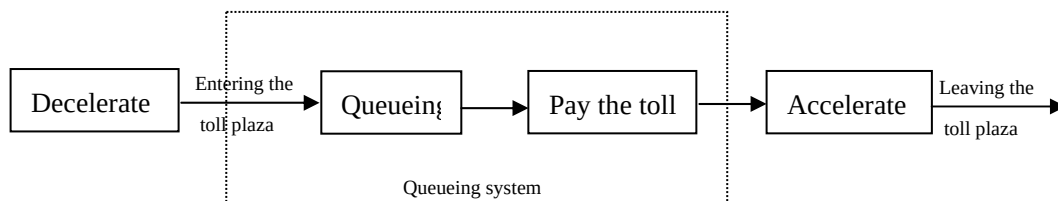


Figure 2

The Process of Paying Toll

According to the Queueing Theory, the course that vehicles wait for paying the toll in line can be considered as a queueing system. In general, a queueing system is constitutive of three basic parts: Input source, Queue discipline and Service facility. In this system, the characteristics of the three parts are discussed next.

Input Source

It is obvious that the input source in this system is the traffic flow. Considering the number of arriving vehicles in a toll plaza, generally, it has characters as follows

- › Let $X(t)$ represents the vehicles that reach the toll plaza in the interval of time $[0, t)$, so that $X(t)$ is independent, that is, without aftereffect.
- › If Δt is small, the probability of one arriving car in $[t, t + \Delta t)$ is independent of t .
- › The probability of two or more arriving cars can be negligible in comparison with the probability of one arriving car in a very short interval of time.

If these three conditions are met, the arrivals of vehicles is considered to be a Poisson process, and the number of arriving vehicles in a toll plaza per time period obeys the Poisson distribution, which has a Probability Function as

$$P\{X(t) = k\} = \frac{\lambda_1^k}{k!} e^{-\lambda_1} \quad \text{for } k = 0, 1, 2, \dots; \lambda_1 > 0$$

where λ_1 is arriving rate, that is the mean number of arriving vehicles per time period.

Queue Discipline

Commonly, a much larger number of tollbooths is provided than the number of travel lanes entering the toll plaza. Consequently, when vehicles spread out through the incoming lanes, some of them have to change lanes. Assume that there are no vehicle-type limits on toll lanes. Since the service priority is First Come First Serve (FCFS), in order to reduce the waiting time in queueing and pass the toll plaza as quickly as possible, motorists always choose the lane on which length of the queue is the shortest.

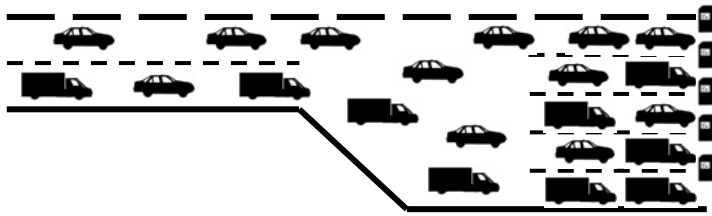


Figure 3
Queueing in Toll Plaza

Service Facility

By the definition of Queueing Theory, the Service Facilities of this system are the tollbooths, and the service time is time for motorists to pay for the toll. The mode of toll collection is clearly of great importance to the service time. Typically, there are four mainly types of toll collection methods in current practice, and they are

- (a) Full service,
- (b) Exactly change or tokens only,
- (c) Tokens
- (d) E-Z Pass.

In general, the service time at the different servers are different, but they obey the negative exponential distribution [J.P.Brans, 1984], which has a Probability Function as

$$F_u(t) = 1 - e^{-\lambda_2 t}$$

where u is the service time, and λ_2 represents the mean Service rate of each tollbooth, that is the number of vehicles served in unit time.

Queueing Model

During the off peak hour the design issue of toll plaza is not critical. Drivers long before arriving at the toll plaza start to search for their appropriate booth and, because the traffic is light, they can easily adjust themselves and make sure that they get to the tollbooth they want to pay tolls, and experience very

minimal delays. Typically the wait in such situations is only one or two cars before paying toll.

During the rush hours, long delays build up at the tollbooth. It almost never happens that some of the booths are idling, while the other long line waiting. This shows that the aggregated departure process from the tollbooth. Since the average delay of a vehicle is not affected by the design of the layout of the different toll collection booths, we assume that there is only one type of payment in a plaza.

The process that the vehicles pass the toll plaza can be divided into two phases:

- › Phase 1: The entry to the toll plaza
- › Phase 2: The departure from the toll plaza

Delay while Paying Tolls

When entering the toll plaza, vehicles decelerate. On the contrary, when leaving the tollbooths, vehicles accelerate. These two courses have hardly any relation to the service of the booths, but relate to the performance of the vehicles and the behavior of the drivers. However, the statuses of queueing and being served have close connection to the service of the booths, and they are the keys to analyze the vehicles delay and the traffic flow capacity.

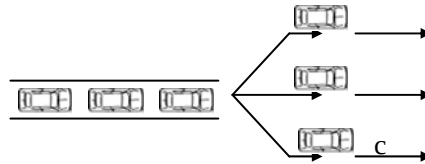


Figure 4

An M/M/c Queueing System

As indicated before, the arrival of vehicles is a Poisson process and the service time obeys the negative exponential distribution. Suppose that the vehicles will go forward after being served, it is a standard M / M / c queueing system according to the Queueing Theory [Hall.R, 1991]. Based on Little's formula, we obtain the average delay as

$$W_q = \frac{L_q}{\lambda_1}$$

where λ_1 is arriving rate of vehicles, and L_q is the length of the queue, which satisfies

$$L_q = \frac{(\lambda_1 / \lambda_2)^n \cdot (\lambda_1 / n\lambda_2)}{n! [1 - (\lambda_1 / n\lambda_2)]^2} \cdot \left[\sum_{k=0}^{n-1} \frac{1}{k!} \left(\frac{\lambda_1}{\lambda_2} \right)^k + \frac{1}{n!} \cdot \frac{1}{1 - (\lambda_1 / n\lambda_2)} \right] (\lambda_1 / \lambda_2)^n$$

where λ_2 represents the mean Service rate of each tollbooth, and n is the number of tollbooths.

Delay while leaving toll plaza

When the flow of vehicles leaves the tollbooths, it squeezes back down to the number of lanes equal to the number of travel lanes before the toll plaza. Hence, a large number of cars are forced to change lanes due to the fewer exits. This is a merging problem, and we apply Queueing Theory to solve this problem by considering each lane in the exit as a server. Firstly, we must calculate the service capacity of the exit.

We calculate the Base Capacity (BC) of a highway for a given Free Flow Speed (FFS), and the following equations were developed based on this information:

$$BC = 1,700 + 10FFS; \text{ for } FFS \leq 70 \text{ mph}$$

$$BC = 2,400; \text{ for } FFS > 70 \text{ mph}$$

Assume that the highway FFS for the highways in the exit is confirmed by the passing-velocity. Applying the Kinematics Formula, we estimate the passing-velocity without lane-changing as

$$v^2 = 2as$$

where v is the velocity of the vehicles, a is the acceleration, and s represents the distance between the tollbooths and the exit.

Then we estimate the acceleration of vehicles. Most of cars claim that “They can accelerate to 60 mph from quiescence in 10 seconds” [Edwards and Hamson, 1989], so the acceleration of vehicles is approximately 2.68 m/s^2 . We understate the acceleration of normal vehicles to be 2.5 m/s^2 . The distance between the tollbooths and the exit is designed as criterion. We assume that the distance s is 100 meters in normal condition. Therefore, we can calculate the passing-velocity without considering lane-changing is 50 miles per hour.

Considering the lane changing in the traffic flow, we should take the interchange density into account. Adjustment Factor for Interchange Density f_{ID} is listed as follows:

Table 2
Interchange Density Adjustment for Free Flow Speed Computation

Interchange per Mile	Reduction in Free-Flow Speed, f_{ID} (mi/h)
0.50	0.0
0.75	1.3
1.00	2.5
1.25	3.7
1.50	5.0
1.75	6.3
2.00	7.5

As the traffic flow is the merging constrainedly in the exit, we assume 2 interchanges per mile. As a result, the reduction in free flow speed f_{ID} is 7.5 miles per hour, and FFS equals to $v - f_{ID}$ [HCM, 2000], so the Freeway Flow Speed is 42.5 mph. Therefore, the Base Capacity of the lanes in the exit is 2125

passenger cars per hour per lane (pcphpl).

The capacity of the segment, in terms of total vehicles per hour (vph), must be computed for a variety of analytic purposes. A few factors are used to adjust Base Capacity. The Peak Capacity is calculated as [HCM, 2000]

$$PC = BC * PHF * N * f_{HV} * f_p$$

In terms of the adjustment factors:

PHF: peak hour factor

N: the number of lanes in one direction

f_{HV} : adjustment factor for heavy vehicles

f_p : adjustment factor for drive populations

The Peak Hour Factor (PHV) is used to account for variations in flow within the peak hour, and defaults of 0.92 for urban facilities and 0.88 for rural facilities [HCM, 2000]. So we take 0.92 in our model.

Adjustment factor for heavy vehicles (f_{HV}) is based on calculating passenger-car equivalents for trucks and buses:

$$f_{HV} = \frac{1}{1 + P_T (E_T - 1)}$$

P_T : proportion of trucks and buses in the traffic stream

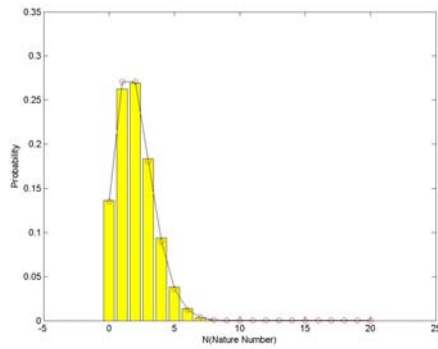
E_T : passenger-car equivalents; 1.5 for all urban highways

We assume that approximately 20 percent expected to be trucks and buses [van Dijk, Hermans and Teunisse, 1999], so P_T is 0.2 in our model.

Adjustment factor for driver population (f_p) is set to 1.0 to indicate that drivers are familiar with roadway and traffic conditions for urban highways [HCM, 2000].

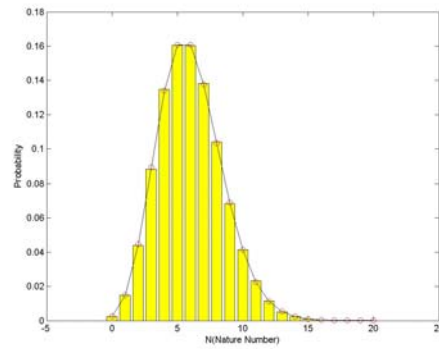
In our model, we consider the Peak Capacity equals to Service Capacity. So, we can get the Service Capacity is approximately $1777N$ cars per hour. Without loss of generality, we also consider that it obeys the negative exponential distribution.

Another important aspect is the input source. In order to search the most efficient of the plaza system, we assume that the traffic flow before the tollbooths is continuous and there is at least one car in the queue all the time. Therefore, we can regard each tollbooth service rate as independent Poisson source with the same parameter λ_2 . As each tollbooth service rate is independent and identically distributed, the arriving rate is $n\lambda_2$ for the phase 2.



(a)

Poisson distribution



(b)

Sum of 3 Poisson distribution

Figure 5

Sum of the three Independent and Identically Distributed Poisson Process

The Queueing system

To sum up, the second step is another standard M / M / c model, so we can calculate the evaluation index in the same way above:

$$W'_q = \frac{L'_q}{n\lambda_2}$$

where λ_2 is mean service rate of each tollbooth, and L'_q is the length of the queue, which satisfies

$$L'_q = \frac{(n\lambda_2 / \lambda_3)^N \cdot (n\lambda_2 / N\lambda_3)}{N! [1 - (n\lambda_2 / N\lambda_3)]^2} \cdot \left[\sum_{k=0}^{N-1} \frac{1}{k!} \left(\frac{n\lambda_2}{\lambda_3} \right)^k + \frac{1}{N!} \cdot \frac{1}{1 - (n\lambda_2 / N\lambda_3)} (n\lambda_2 / \lambda_3)^N \right]^{-1}$$

where λ_3 is the equivalent service rate of each lane in the exit, and N is the number of lanes.

Traffic Flow Model

Above, we have studied the problem of toll plaza design, based on Queueing Theory. Inside, we find some limitations in our model. For example, when the arriving rate is larger than the average service rate, queue model can't solve the problem. However, it is the key problem, that is, most of traffic congestions usually takes place in the condition.

We analyze some situations, and find that when the entrance arriving rate is larger than the exit overall Peak Capacity, the method based on Queueing Theory can't work, but it is the very time when congestions take place easily. In rush hour, we focus on how the traffic flow can approach the maximum capacity as possible. In this situation, our optimal goal has been changed into how to get the maximum traffic flow in the toll plaza.

Comparing the traffic flow to fluid, Lighthill and Whitham research the law of the traffic flow in the ease of high traffic flow density, and put forward the Traffic Flow Theory which is also called L-W theory as below:

$$\frac{\partial \rho(x,t)}{\partial t} + \frac{\partial f(x,t)}{\partial x} = 0$$

$$f(x,t) = \rho(x,t)v(x,t)$$

where $f(x,t)$ is the traffic flow, $\rho(x,t)$ is the traffic flow density, $v(x,t)$ is the velocity of the traffic flow, and x, t represent the distance and time index.

In order to research the relationship between the speed and the density, Grenberg derived a logarithm model based on the L-W theory, which can be used in the condition of high traffic flow density:

$$v = v_m \ln \frac{\rho_m}{\rho}$$

where v_m is the critical velocity, ρ_m is the jam-density. Thus, we have

$$f = v \rho_m e^{-\frac{v}{v_m}}$$

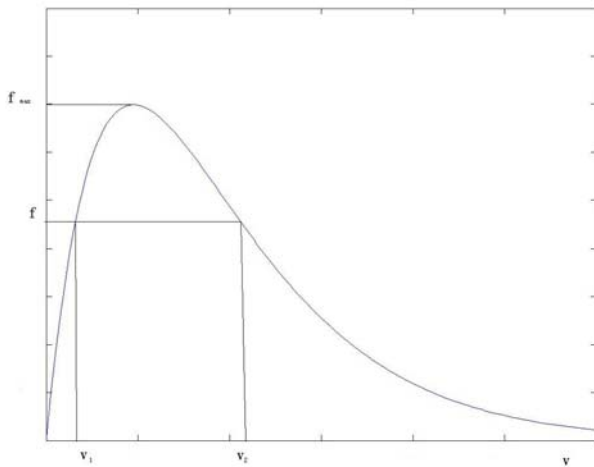


Figure 6
Flow vs. Velocity

The speed-flow diagram is the envelope of all possible combinations of the effective speed and traffic flow. In Figure 6, it is also illustrated that although every speed v corresponds with one unique traffic flow f , the reverse is not true. There are two speeds for every traffic flow: an upper branch (v_2) where speed decreases with flow and a lower branch (v_1) with an increasing speed in terms of flow. An intuitive explanation can be as follows: as the flow moves from v_m to $f_{\max}$, congestion increases but the flow rises because the decline in speed is offset by the higher volume. If traffic continues to enter the flow is larger $f_{\max}$, flow falls because the decline in speed more than offsets the additional vehicle numbers further increasing congestion [Daganzo, 1997].

Considering the toll plaza system, the traffic capacity of this system is due to the capacity of lanes in the exit. So we confirm the number of tollbooths according to the exit Peak Capacity when the entrance arriving rate is larger than the exit overall Peak Capacity, that is, $n\lambda_2$ should approach to the Peak capacity in the exit as much as possible.

The Optimal Number of Tollbooths

In a word, we can use our two models for different toll plazas design with different optimal goals:

When the entrance arriving rate is smaller than the exit overall Peak Capacity, our optimal goal is to minimize the average delay T_{delay} for both two queueing processes.

$$\begin{aligned} \min \quad & T_{delay} = W_q + W'_q \\ \text{subject to} \quad & \begin{cases} \lambda_1 < n\lambda_2 \\ n\lambda_2 < N\lambda_3 \end{cases} \end{aligned}$$

When the entrance arriving rate is larger than the exit overall Peak Capacity, our optimal goal is to maximize traffic flow f , that is, $n\lambda_2$ should approach to the Peak capacity $N\lambda_3$ in the exit as much as possible.

Performance Analysis

To validate our model, we search for traffic data in some plazas in Garden State Parkway. Average Annual daily Traffic (AADT) is important data for estimating the arriving rate in the plazas. Data from the continuous traffic count stations above were used to examine the fluctuation of traffic volumes by time of day (see Figure 7) plots the average volume by time of day for an average 24-hour period for 4 toll plazas. Figure 7 clearly illustrates the morning and afternoon peaks in traffic volumes. These two rush hours should be the major causation that can lead traffic congestion, especially in the toll plazas.

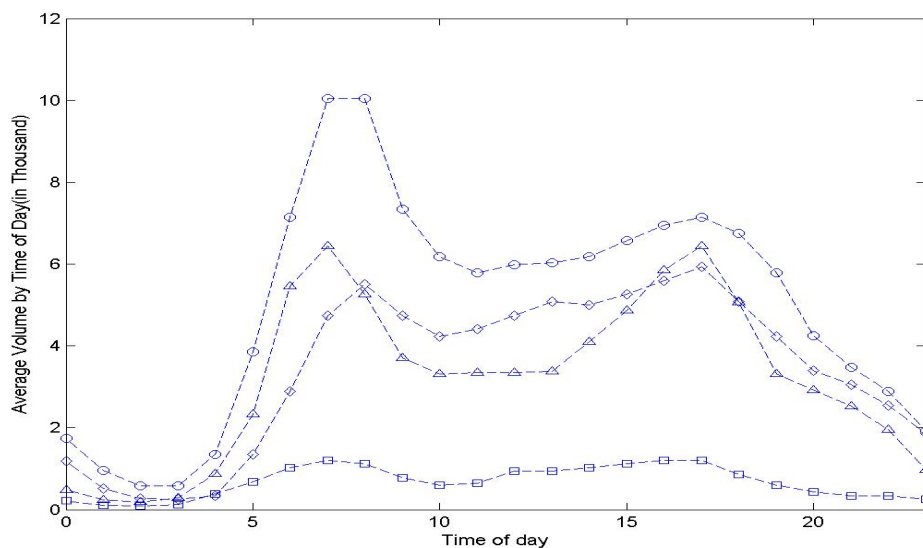


Figure 7 The

Average Volume by Time of Day

We estimate the service time based on the different vehicles and different service payment styles. Four popular payment styles are: Cash, Debit Card, Credit Card, and E-Z pass. The average service time for each style is listed as follows:

Table 3

Average Service Times per Vehicle Type (in seconds).

[van Dijk, Hermans and Teunisse, 1999]

	Cars	Trucks
E-Z pass	3.8	5.1
Cash	7.5	11.3
Debit Card	20.0	23.0
Credit Card	4.5	10.1

The percentage of drivers expected to use a certain payment system is therefore an important parameter:

Table 4

Distribution of Preferred Payment

Systems during a Tourist Peak Hour. (%) [van Dijk, Hermans and Teunisse, 1999]

	Cars	Trucks
E-Z pass	55	70
Cash	25	15
Debit Card	11.25	7.5
Credit Card	11.25	7.5

The arriving vehicles' pattern is also an important parameter, listing as follows. For simplification, the percentage of cars is 80%, while the percentage of the trucks is 20%.

Table 5

Number of Vehicles Arriving per Hour in the Direction North-South

[van Dijk, Hermans and Teunisse, 1999]

	Non-tourist Peak Hour	Tourist Peak Hour
Cars	370	436
Trucks	103	103
Total	473	539

According to the percentage and service time of different payment patterns, we can calculate the average service rate 0.1531 cars per second

We take two representative plazas into our models: Cape May Toll Plaza and Union Toll Plaza.

Cape May Toll Plaza is not so crowded. The AADT is 16,207 cars per day in north direction. Take 7.5% of AADT for a rush hour's flow. We calculate the $\lambda_1 = 0.3377$ cars per second (cps). $\lambda_2 = 0.1531$ cars per second. The number of incoming lanes N is 2. Apparently,

$$\lambda_1 < N\lambda_2 = \frac{1777 * 2}{3600}$$

so we can use Queueing model to identify the optimal number of tollbooths. Applying the program

designed in Matlab (In Appendix A), we get the optimal number of tollbooths is 3 for Cape May Toll Plaza in northern direction.

Unlike Cape May, Union Toll Plaza is a one of the most congested plaza in the Garden State Parkway. The AADT is 101,724 cars per day in northern direction. Take 7.5% of AADT for an hour in rush hour. We calculate the $\lambda_1 = 2.1193$ cars per second. $\lambda_2 = 0.1531$ cars per second. The number of incoming lanes N is 3. Apparently,

$$\lambda_1 > N\lambda_3 = \frac{1777 * 3}{3600}$$

so we can't use Queueing Model any longer. To identify the optimal number of tollbooths, we use Traffic Flow Model. We estimate the optimal number of tollbooths is 12 for Union Toll Plaza in northern direction.

Table 6

The Result of the Model

	λ_1 (cps)	λ_2 (cps)	N(lane)	Suitable model	Optimal tollbooths	Actual tollbooths
Cape May Toll Plaza	0.3377	0.1531	2	Queueing Model	3	5
Union Toll Plaza	2.1193	0.1531	3	Traffic Flow Model	12	13(future)

In Cape May Toll Plaza, the optimal number of tollbooths is smaller than the actual number. Because this plaza is not so crowded, the style of payment may be chosen less effectively considering the expense of the tollbooths. While in Union Toll Plaza, a great number of cars flood into the plaza in rush hour, the number of tollbooths can't meet the requirement. So 13 tollbooths are used in the March, 2005. Based on our model, the congestion could be relaxed.

A simulation program is designed to validate our model (In Appendix B).

Further Discussion

Mixed-type tollbooths

The current practice in Garden State Parkway is that the four payment types of tollbooths are mixed in most toll plazas. At the toll plaza, arriving vehicles have to choose a tollbooth depending on their payment system and the queueing length at the tollbooths that offered the desired payment system. Four popular payment styles are: Cash, Debit Card, Credit Card, and E-Z pass. E-Z pass is considered as the fast channels for cars.

One option would be to separate the lanes and to offer only one type of payment system at every tollbooth (see Figure 7a). The disadvantage of a separate lane system is that certain tollbooths can be underutilized while others will be overloaded. This can be inefficient in operational usage. Analytical

queueing systems can predict the behavior of separate lane queueing system well.

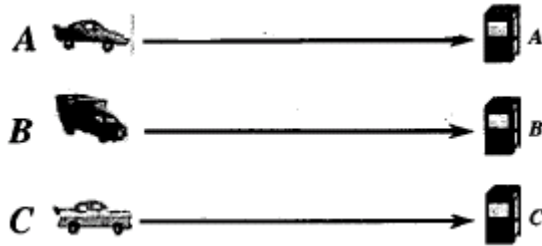


Figure 7a

A Separate Lane System

Another option would be to offer all payment systems at all tollbooths and have the vehicles queue in one line (see Figure 7b). Each tollbooth would accept cash, credit cards, and electronic payment. The advantage in this case is that it provides more operational flexibility while it also increases the efficiency of the tollbooths. The disadvantage, however, is that the wide range of service times introduces variability. As variability in service times is one of the main causes for queues to arise, this disadvantage can dominate the gain in efficiency. A more detailed quantitative study taking into account the actual numbers is therefore required.

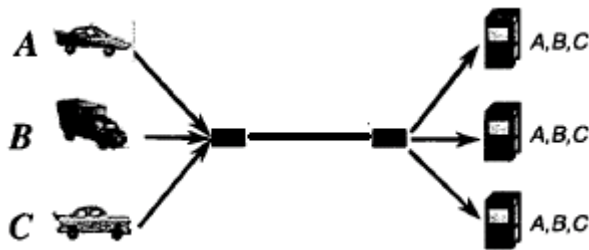


Figure 7b

An all payment systems

To calculate the number of each type of service, we can use simulation method. An important part of the simulation study was to provide valid input data for the simulation model, such as traffic forecasts, service times, and parking times. Given the direct relation between the input data and the performance of the toll system, it was important that the input data and assumptions were accurate. All the information should be collected.

Change of Toll System Types

As indicated before, when the traffic is busy, it has negative effects to the exit of the plaza while increasing the number of tollbooths. Contrarily, decrease the tollbooths may leads to congestion problems at the entry to the toll plaza. Hence, in this condition, things may go smoothly if change the toll system types properly. Based on the current practice, we propose two improved methods in designing of a toll plaza system.

Due to the assumption, we only discussed the scenario where there is exactly one tollbooth per incoming lane. A new method is that two tollbooths are setting in tandem on each lane, which is shown in Figure

8.

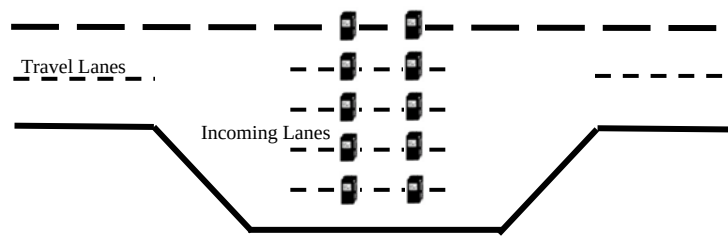


Figure 8
Tollbooths in Tandem

In this system, two cars are served simultaneously on each toll lane. Consequently, the traffic capacity is significantly improved [Xiaonv Lou, 2000]. This method can be used in the section of a highway where the traffic is extremely heavy. Nevertheless, this will prolong the delay time, which is a limitation.

Consider the other method shown in Figure 9. Since the people on and off duty drive on the highway, which result in two peak traffic flows. As the direction of the two flows is opposite, the central part of lanes can turn into lanes in common used. For instance, if people on duty drive from west to east, the common used lanes are turn into the direction from west to east, vice versa.

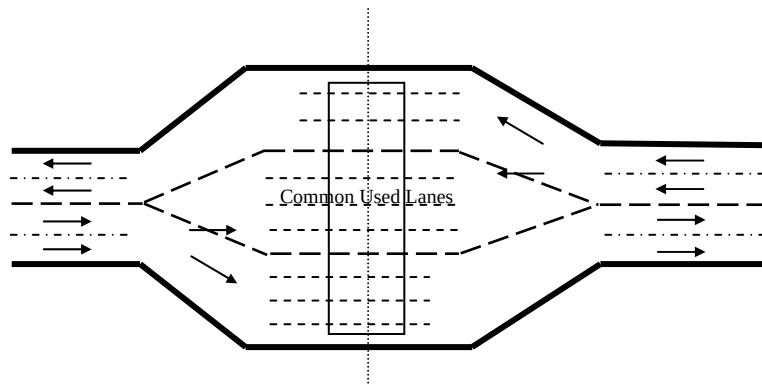


Figure 9
Toll Plaza with Common Used Lanes

Strengths and Weaknesses

Analyze the traffic flow passing through the plaza in two conditions. Therefore, we can solve most of problems under different conditions, and give the optimal toll plaza design.

Comparison with the present strategies. In reality, we find several tollbooths are not disposed effectively. According to our model, we propose the improved advices, such as the Cape May Toll Plaza on I 9-2-101.

Ease of use. Although the concepts behind our model are relatively difficult to understand, the flow chart is simple to perform, and then we designed a program that allowed different data to input to simulate the waiting process.

The weakness is that the simplification and approximation taken in the process of going through toll plazas introduce errors.

References

Garden State Parkway. <http://www.nycroads.com/roads/garden-state.html>. Accessed 5 February 2005.

Freeway Capacity. <http://www.fhwa.dot.gov/ohim/hpmsmanl/appn1.htm>. Accessed 6 February 2005.

Nico M. van Dijk, Mark D. Hermans, Maurice J.G. Teunisse. 1999. Designing the Wester Schelde Tunnel Toll Plaza Using A Combination of Queueing and Simulation. Netherland: University of Amsterdam.

Xiuli Chao, Ph.D. 2002. Design and Evaluation of Toll Plaza Systems. New Jersey Institute of Technology. Newark.

J.P.Brans. 1984. Operational Research. North Holland: North Holland Publish Company.

Hall, R. (1991). Queueing Methods for Service and Manufacturing, Prentice-Hall, New Jersey.

Dilwyn Edwards, Mike Hamson. 1989. Guide to Mathematical Modeling. London: Macmillan Education LTD.

Daganzo, C.F., 1997, Fundamentals of Transportation and Traffic Operations, Elsevier Science Ltd., Oxford

Appendix A

The search program to determine the optimal number

```
function r=Optimal
% compute the model
% find the best design of the tollbooths
% all the formulas are based on our model
% written on 7th, Feb, 2005
for desk_num=1:40
% the serving queue model
```

```

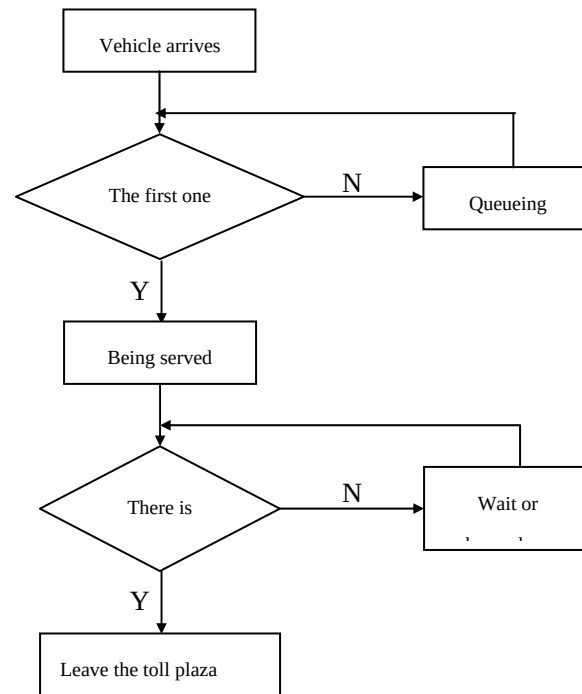
lane_num=2;
lamda_in=0.3376;
lamda_ser=0.1464;
lamda_out=0.4936;
tempsum=0;
fac=factorial(desk_num);
lamratio=lamda_in/lamda_ser;
for ser=0:(desk_num-1)
tempsum=tempsum+(1/factorial(ser))*(lamratio)^ser;
end % for
prob_ser=tempsum+(1/fac)*(1/(1-lamratio/desk_num))*(lamratio^desk_num);
prob_ser=1/prob_ser;
queue_ser=(lamratio^desk_num)*(lamratio/desk_num)*prob_ser/(fac*(1-lamratio/desk_num)^2);

% the squeezing out model
laneratio=desk_num/lane_num;
outratio=lamda_ser/lamda_out;
fac1=factorial(lane_num);
tempsum1=0;
for out=0:(lane_num-1)
tempsum1=tempsum1+(1/factorial(out))*((desk_num*outratio)^out);
end % for
prob_out=tempsum1+(1/fac1)*(1/(1-laneratio*outratio))*((desk_num*outratio)^lane_num);
prob_out=1/prob_out;
queue_out=((desk_num*outratio)^lane_num)*(laneratio*outratio)*prob_out/(fac1*((1-laneratio*outratio)^2));
time_ser=queue_ser/lamda_in; % average waiting time while being served
time_out=queue_out/(desk_num*lamda_ser); % average waiting time while squeezing out
totaltime=time_ser+time_out;
if (lamratio<desk_num & (lane_num/outratio)>desk_num)
if (queue_ser>=1)
disp('Platform number:'),disp(desk_num)
disp('Waiting time:'),disp(totaltime)
else
disp('Can not meet the queue length!')
end % if
else
disp('Not the probable lane!')
end
end

```

Appendix B

Simulation program



```

function r=ToolSimulate
% simulate the cars arriving toll plazas and leaving
% compute the average waiting time and the longest waiting time
% simulate N cars in a rush hour
% written on 8th, Feb, 2005
%N=6000;
lane=7;
way=3;
traffic=0.3376/lane; % the traffic flow, refer to I-9-2-106, 2003
N=ceil(3600*lane*traffic);
t0=0.1; % two cars keep time interval
bigSer=0.1247; % service ability for trucks
smaSer=0.1531; % service ability for mincars
bigOut=0.2800; % efficiency of trucks to go out
smaOut=0.3200; % efficiency of mincars to go out
carsum=0;
for k=1:N
    ktest=rand;
    if (ktest>=0 & ktest<=1/lane)
        carsum=carsum+1;
    end % if
  
```

```

end % for
waittime=zeros(1,carsum); % initialize the matrix of waiting time
usetime=zeros(1,carsum); % initialize the matrix of servering time
for num=1:carsum
if (num==1)
waittime(num)=0; % the first car
cartest=rand;
if (cartest>=0 & cartest<0.8) % mincar counts 80 percent
rtest=rand;
usetime(num)= (-1/smaSer)*log(rtest); % exponential distribution
else
rtest=rand;
usetime(num)= (-1/bigSer)*log(rtest);
end % if
else
gtest=rand;
deltatime=( -1/traffic)*log(gtest)-t0; % poisson flow
waittime(num)=waittime(num-1)+usetime(num-1)-deltatime; % refer to the classical formula
if (waittime(num)<0)
waittime(num)=0;
end % if
cartest=rand;
if (cartest>=0 & cartest<0.8) % mincar counts 80 percent
rtest=rand;
usetime(num)= (-1/smaSer)*log(rtest);
else
rtest=rand;
usetime(num)= (-1/bigSer)*log(rtest);
end % if
end % if
end % for

outsum=0;
for m=1:N
mtest=rand;
if (mtest>=0 & mtest<1/way)
outsum=outsum+1;
end % if
end % for
outwait=zeros(1,outsum); % intialize the matrix of waiting time while squeezing out
outuse=zeros(1,outsum); % intialize the matrix of using time while squeezing out
for out=1:outsum
if (out==1)

```

```
outwait(out)=0; % free lane to use
vehetest=rand;
if (vehetest>=0 & vehetest<0.8)
rtest=rand;
outuse(out)=(-1/smaOut)*log(rtest);
else
rtest=rand;
outuse(out)=(-1/bigOut)*log(rtest);
end % if
else % if
vehetest=rand;
if (vehetest>=0 & vehetest<0.8)
rtest=rand;
outdelta=(-1/(lane*smaSer/way))*log(rtest); % the random time interval
else
rtest=rand;
outdelta=(-1/(lane*bigSer/way))*log(rtest); % the random time interval
end % if
outwait(out)=outwait(out-1)+outuse(out-1)-outdelta;
if (outwait(out)<0)
outwait(out)=0;
end % if
vehetest=rand;
if (vehetest>=0 & vehetest<0.8)
rtest=rand;
outuse(out)=(-1/smaOut)*log(rtest); % exponential distribution
else
rtest=rand;
outuse(out)=(-1/bigOut)*log(rtest);
end % if
end % if
end % for

disp('Average waiting time:'),disp((sum(waittime)/N)+(sum(outwait))/N)
```